

Wreath Product Symmetry and Permutation Alignment for Transformer Task Vector Transport

CSMATH 2026 Course Paper

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Abstract

When foundation models are updated, fine-tuned downstream models risk obsolescence—retraining is expensive and original training data is often unavailable. This paper addresses the question: can fine-tuning knowledge, encoded as a *task vector* (weight difference), be safely transported to a new model release without data or retraining? We provide a group-theoretic formalization. Under the head-block-preserving constraint, we prove that the maximal permutation symmetry subgroup of multi-head attention (MHA) is the wreath product $\mathcal{W}(H, d_k) = \mathcal{S}_H \wr \mathcal{S}_{d_k}$ (Theorem 1), providing a group-theoretic account of previously ad-hoc two-level alignment procedures. We further prove that the alignment optimization decomposes into an outer linear assignment problem (LAP) whose cost matrix is given by H^2 inner LAPs (Theorem 2), yielding the global optimum under the wreath-product constraint—in contrast to heuristic two-stage schemes using spectral distance as the outer cost. We establish spectral proxy consistency and Weyl-inequality bounds (Theorem 3), with finite-step convergence of the multi-layer alternating optimization (Theorem 4). Three experiments examine the theory: exhaustive enumeration confirming the wreath product characterization (Experiment 1), comparison of exact decomposition against spectral proxy on alignment scores (Experiment 2), and small-scale proof-of-concept task vector transport on TinyViT (Experiment 3).

Keywords: computational applied mathematics; wreath product; multi-head attention; permutation alignment; task vector transport; linear assignment problem

1 Introduction

1.1 Background and Motivation

Foundation models such as LLaMA and Qwen release new versions every few months. Downstream users fine-tune these models for specific tasks (domain classification, instruction following, preference alignment). When a new base model is released, the fine-tuned downstream model becomes *stale*: directly using the old model forgoes improvements in the new release, while retraining on the new base is expensive and often impossible because the original training data is unavailable due to privacy or storage constraints.

A promising approach uses *task vectors* [Ilharco et al.(2023)Ilharco, Ribeiro, Wortsman, Gururangan, Schmidt, Hajishirzi, and Farhadi]: encode fine-tuning knowledge as the weight difference $\tau = \theta_{\text{ft}} - \theta_{\text{base}}$, then “add” it to the new model: $\theta_{\text{new}} + \alpha \cdot \tau$. However, this arithmetic relies on an implicit assumption: corresponding neurons in the old and new models share the same *semantics*. Due to permutation symmetries in neural network training—arbitrarily permuting hidden neurons and adjusting subsequent layers yields a functionally equivalent network—two independently trained models may have completely different internal neuron orderings. This is the core challenge of the *model rebasin* problem [Ainsworth et al.(2023)Ainsworth, Hayase, and Srinivasa, Entezari et al.(2022)Entezari, Sedghi, Saukh, and Neyshabur].

We therefore ask: **given the permutation symmetry group of the architecture, can we find the correct alignment to safely transport a task vector from an old model to a new one?**

1.2 Related Work

Task vectors and model editing. Ilharco et al. [Ilharco et al.(2023)Ilharco, Ribeiro, Wortsman, Gururangan, Schmidt, Hajishirzi, and Farhadi] introduced task vectors, demonstrating weight-space arithmetic for editing model behavior. Its effectiveness, however, depends on weight-space “semantic linearity”—when neuron alignment is disrupted by permutations, the arithmetic breaks down.

Model re-basin and Git Re-Basin. Ainsworth et al. [Ainsworth et al.(2023)Ainsworth, Hayase, and Srinivasa] proposed Git Re-Basin, aligning two independently trained models via permutation matching. Their method solves linear assignment problems (LAPs) on flattened weight matrices, which works well for MLPs and CNNs. However, flattening ignores the *head structure* of multi-head attention: each head computes its own independent softmax, so mixing dimensions across heads destroys the attention mechanism’s functional semantics. Entezari et al. [Entezari et al.(2022)Entezari, Sedghi, Saukh, and Neyshabur] and Singh et al. [Singh and Jaggi(2023)] studied permutation symmetries in deep networks theoretically, but did not specifically address MHA structural constraints.

TransFusion. Rinaldi et al. [Rinaldi et al.(2025)Rinaldi, Capitani, Bonicelli, Crisostomi, Bolelli, Ficarra, Rodolà, Calderara, and Porrello] proposed TransFusion, which adapts the Git Re-Basin framework specifically for Transformer task-vector re-basin. Its core innovation is a two-level alignment: first matching heads via singular-value spectral distance, then matching dimensions within each head. The method empirically demonstrates task vector transport. However, the two-level procedure remains *heuristic* in its theoretical grounding: it does not characterize the symmetry structure in group-theoretic terms, nor does it analyze the optimality of the two-stage decomposition.

Connection to course material. The mathematical framework of this paper connects to multiple CSMATH topics: LAPs as linear programming on the Birkhoff polytope (linear programming/duality), SVD and spectral analysis (multivariate statistics/PCA), wreath product group actions (applied functional analysis), and alternating optimization convergence (nonlinear optimization/coordinate descent).

1.3 Contributions

This paper provides a group-theoretic formalization of two-level Transformer alignment. Our primary contribution is *re-expressing* the previously heuristic two-level procedure in the language of wreath products, and identifying a specific algorithmic improvement (exact outer cost). Concretely:

- (1) **Symmetry group characterization** (Theorem 1): under the head-block-preserving constraint, we identify the maximal permutation symmetry subgroup of MHA as the wreath product $\mathcal{W}(H, d_k) = \mathcal{S}_H \wr \mathcal{S}_{d_k}$.
- (2) **Exact decomposition** (Theorem 2): we prove that the alignment problem decomposes into a single outer $H \times H$ LAP whose cost matrix is determined by H^2 inner $d_k \times d_k$ LAPs, yielding the global optimum under the wreath constraint—a different mathematical object from heuristic schemes using spectral distance as outer cost.
- (3) **Spectral proxy analysis** (Theorem 3): we establish invariance of the SVD-based spectral distance under intra-head permutations and its lower-bound relationship to the exact cost via Weyl’s inequality.
- (4) **Convergence guarantee** (Theorem 4): we prove monotonic convergence of alternating optimization over the finite wreath product group.

- (5) **Experimental examination:** three experiments check the theory’s verifiability—exhaustive enumeration of wreath symmetries (Exp. 1), alignment score comparison of exact vs. spectral (Exp. 2), and small-scale proof-of-concept task vector transport (Exp. 3).

We do *not* claim a new algorithmic paradigm; the wreath product structure and two-level alignment have been implicitly used in prior work such as TransFusion [Rinaldi et al.(2025)Rinaldi, Capitani, Bonicelli, Crisostomi, Bolelli, Ficarra, Rodolà, Calderara, and Porrello]. Our contribution is the group-theoretic characterization and the identification of an improved (albeit computationally costlier) variant of the outer cost.

1.4 Paper Structure

Section 2 formalizes the problem and introduces the wreath product group. Section 3 develops geometric intuition. Section 4 presents algorithms and Theorems 1–4 with proofs. Section 5 reports three experiments. Section 6 discusses strengths, limitations, and connections to course material.

2 Problem Formalization

2.1 Notation and Setup

Consider an L -layer Transformer $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ with parameters $\theta = \{W_\ell\}_{\ell=1}^L$. Let θ_A be the original model fine-tuned on a downstream task, θ_B the new release (same architecture, different training), and θ_A^{ft} the fine-tuned version. The *task vector* is $\tau = \theta_A^{\text{ft}} - \theta_A$.

Multi-head attention structure. For an MHA layer with embedding dimension d_m , H heads, and per-head dimension $d_k = d_m/H$, the projection matrices $W_Q, W_K, W_V \in \mathbb{R}^{d_m \times d_m}$ admit row-wise partitioning:

$$W_Q = \begin{bmatrix} W_Q^{(1)} \\ \vdots \\ W_Q^{(H)} \end{bmatrix}, \quad W_K = \begin{bmatrix} W_K^{(1)} \\ \vdots \\ W_K^{(H)} \end{bmatrix}, \quad W_V = \begin{bmatrix} W_V^{(1)} \\ \vdots \\ W_V^{(H)} \end{bmatrix}, \quad (1)$$

where $W_Q^{(h)}, W_K^{(h)}, W_V^{(h)} \in \mathbb{R}^{d_k \times d_m}$ correspond to head h . For input $X \in \mathbb{R}^{n \times d_m}$ (n tokens), head h computes:

$$Q_h = X(W_Q^{(h)})^\top, \quad K_h = X(W_K^{(h)})^\top, \quad V_h = X(W_V^{(h)})^\top, \quad O_h = \text{softmax}\left(\frac{Q_h K_h^\top}{\sqrt{d_k}}\right) V_h, \quad (2)$$

with final output $O = [O_1 \mid \dots \mid O_H] \cdot W_O \in \mathbb{R}^{n \times d_m}$.

Permutation action. A permutation matrix $P \in \mathcal{S}_N$ satisfies $P^\top P = I_N$. In neural networks, P acts on weight matrices via row permutation ($P \cdot W$) or column permutation ($W \cdot P^\top$), reordering neurons.

2.2 The Wreath Product Group

Definition 2.1 (Wreath Product). *For permutation groups $G \subseteq \mathcal{S}_H$ and $K \subseteq \mathcal{S}_{d_k}$, the wreath product $G \wr K$ is*

$$G \wr K = \{(\sigma; \tau_1, \dots, \tau_H) \mid \sigma \in G, \tau_h \in K\}, \quad (3)$$

with group operation $(\sigma; \{\tau_h\}) \circ (\sigma'; \{\tau'_h\}) = (\sigma\sigma'; \{\tau_{\sigma'(h)} \circ \tau'_h\})$. It is isomorphic to the semidirect product $G \ltimes K^H$, of order $|G| \cdot |K|^H$.

Taking $G = \mathcal{S}_H$ (inter-head permutations) and $K = \mathcal{S}_{d_k}$ (intra-head permutations):

$$\boxed{\mathcal{W}(H, d_k) \triangleq \mathcal{S}_H \wr \mathcal{S}_{d_k}, \quad |\mathcal{W}(H, d_k)| = H! \cdot (d_k!)^H.} \quad (4)$$

An element $P = (\sigma; \tau_1, \dots, \tau_H) \in \mathcal{W}(H, d_k)$ acts on MHA projection matrices as

$$P \cdot W_Q^{(h)} = \tau_h \cdot W_Q^{(\sigma^{-1}(h))}, \quad h = 1, \dots, H, \quad (5)$$

and analogously for W_K, W_V . Intuitively: σ reassigns heads to new positions, then each τ_h permutes dimensions within its head.

2.3 Alignment Problem

Given two independently trained MHA layer weights W^A and W^B , we seek $P \in \mathcal{W}(H, d_k)$ maximizing the alignment inner product:

$$P^* = \arg \max_{P \in \mathcal{W}(H, d_k)} \langle W^B, P \cdot W^A \rangle, \quad (6)$$

where $\langle W^B, P \cdot W^A \rangle = \sum_{M \in \{Q, K, V\}} \sum_{h=1}^H \langle W_M^{B, (h)}, \tau_{\sigma^{-1}(h)} \cdot W_M^{A, (\sigma^{-1}(h))} \rangle$.

After alignment, P transports the task vector: $\hat{\theta}_B^{\text{ft}} = \theta_B + \alpha \cdot P(\tau)$, with scaling factor $\alpha > 0$.

Multi-layer extension. For an L -layer Transformer, each layer ℓ has an independent permutation $P_\ell \in \mathcal{W}(H, d_k)$, with the previous layer’s permutation propagating through residual connections:

$$\max_{\{P_\ell\}_{\ell=1}^L} \sum_{\ell=1}^L \langle W_\ell^B, P_\ell \cdot W_\ell^A \cdot P_{\ell-1}^\top \rangle, \quad P_0 = I, \quad P_{L+1} = I. \quad (7)$$

Problem class. The search space $\mathcal{W}(H, d_k)^L$ is a finite group of size $(H! \cdot (d_k!)^H)^L$. Each subproblem (inter-head and intra-head LAP) is convex—LAPs admit LP relaxations on the Birkhoff polytope—but the joint multi-layer optimization (7) is non-convex due to coupling between layers via residual connections. All subproblems are solvable via the Hungarian algorithm.

3 Mathematical Background and Intuition

3.1 Geometric Picture of the Wreath Product

The wreath product $\mathcal{W}(H, d_k) = \mathcal{S}_H \wr \mathcal{S}_{d_k}$ is best understood in two steps.

Step 1: Inter-head permutation $\mathcal{S}_H$. Imagine H containers, each holding d_k neurons. $\mathcal{S}_H$ swaps entire containers—moving container $\sigma^{-1}(h)$ to position h .

Step 2: Intra-head permutation $\mathcal{S}_{d_k}$. Within each container, $\tau_h \in \mathcal{S}_{d_k}$ re-numbers the d_k neurons.

Why must it be a wreath product? Can an arbitrary $\mathcal{S}_{d_m}$ permutation replace the wreath product structure? No—if dimensions from one head are split across multiple heads, the per-head softmax normalization is disrupted, since each head computes its own independent softmax:

$$A_h = \text{softmax} \left(\frac{Q_h K_h^\top}{\sqrt{d_k}} \right) \quad (\text{independent per head}). \quad (8)$$

Mixing dimensions from different heads into a single softmax couples originally independent attention competitions. Thus *preserving head boundaries* requires exactly the wreath product structure. This intuition is made rigorous in Theorem 1.

3.2 Computational Invariance under Permutation

A toy example. Consider MHA with $d_m = 4, H = 2, d_k = 2$. Let $W_Q^{(1)}$ extract the first two dimensions of $\mathbf{x}$ and $W_Q^{(2)}$ extract the last two. Applying the inter-head swap $\sigma = (1 \ 2)$, $\tau_1 = \tau_2 = I_2$ yields: $Q'_1 = (x_3, x_4)$, $Q'_2 = (x_1, x_2)$. The two heads’ Q-values are swapped, but the attention computation within each head is unchanged—the outputs O_1 and O_2 simply exchange positions. The wreath permutation is *transparent* to the MHA computation: it renames dimensions without altering results.

Why $\tau_h^\top \tau_h = I$ is crucial. Under intra-head permutation τ_h , we have $Q'_h = Q_h \tau_h^\top$ and $K'_h = K_h \tau_h^\top$. The attention score is:

$$Q'_h (K'_h)^\top = Q_h \tau_h^\top \tau_h K_h^\top = Q_h K_h^\top,$$

since permutation matrices are orthogonal ($\tau_h^\top \tau_h = I$). The permutation “cancels out,” the attention matrix is unchanged, and $V'_h = V_h \tau_h^\top$ yields $O'_h = O_h \tau_h^\top$. The permutation propagates cleanly to the output, establishing equivariance: $f_{P(W)}(X) = P \cdot f_W(X)$.

3.3 Intuition for Spectral Distance

The SVD of a head weight matrix $W \in \mathbb{R}^{d_k \times d_m}$ is $W = U \Sigma V^\top$, with singular values $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_{d_k})$. Under permutation $\tau \in \mathcal{S}_{d_k}$, $\tau W = (\tau U) \Sigma V^\top$ —only the left singular vectors change; Σ is invariant. Thus $D_{ij} = \|\Sigma_i - \Sigma_j\|_F$ is a *permutation-invariant* distance between heads, making it a natural candidate for inter-head matching. However, by Weyl’s inequality, D_{ij} is only a *lower bound* on $\min_\tau \|W_i - \tau \cdot W_j\|_F$; two heads may share identical singular values but have misaligned singular vectors, in which case the spectral proxy fails to distinguish them. The exact cost c_{ij} in Theorem 2 resolves this by using the true optimal intra-head alignment cost.

3.4 Connection to Course Material

The wreath product analysis connects to *applied functional analysis* (group actions and symmetry). The decomposition into inter-head and intra-head LAPs belongs to *combinatorial optimization* (LAPs as LPs on the Birkhoff polytope). The alternating optimization convergence analysis draws on *non-linear optimization* (coordinate descent). SVD and spectral methods extend *multivariate statistics* (PCA).

4 Method Analysis

4.1 Algorithms

We present two algorithm variants. Algorithm 1 (exact decomposition, Theorem 2) uses optimal intra-head LAP costs for the outer cost matrix, yielding the global optimum under the wreath constraint. Algorithm 2 (spectral proxy, Theorem 3) uses SVD-based spectral distance as the outer cost, requiring only H inner LAPs.

The key difference: Algorithm 1 precomputes all H^2 optimal intra-head alignment costs c_{ij} ; Algorithm 2 substitutes the spectral distance D_{ij} , computing only H intra-head LAPs (post outer matching).

4.2 Inter-Head and Intra-Head Assignment

For the spectral proxy variant, the inter-head matching solves

$$\min_{\sigma \in \mathcal{S}_H} \sum_{i=1}^H D_{i, \sigma(i)}, \quad D_{ij} = \sum_{M \in \{Q, K, V\}} \|\Sigma_i^B(W_M) - \Sigma_j^A(W_M)\|_F, \quad (9)$$

where $\Sigma_i(W_M)$ are the singular values of head i ’s projection matrix. Permutation invariance (Theorem 3) guarantees well-definedness. This $H \times H$ LAP is solved by the Hungarian algorithm in $O(H^3)$ time.

For each matched head pair, the intra-head alignment maximizes the inner product:

$$\tau^* = \arg \max_{\tau \in \mathcal{S}_{d_k}} \sum_{M \in \{Q, K, V\}} \langle W_M^{B, (i)}, \tau \cdot W_M^{A, (\sigma^*(i))} \rangle, \quad (10)$$

also a $d_k \times d_k$ LAP solvable in $O(d_k^3)$. The H intra-head problems are independent.

Algorithm 1: Wreath Alignment (Exact Decomposition)

Input : Two L -layer Transformer weights $\theta_A, \theta_B; d_m, H, d_k$

Output: Per-layer permutations $\{P_\ell\}_{\ell=1}^L \subset \mathcal{W}(H, d_k)$

```
1 Initialize  $P_\ell \leftarrow I_{d_m}$  for all  $\ell$ ;  $P_0, P_{L+1} \leftarrow I_{d_m}$ ;
2 repeat
3   for  $\ell = 1$  to  $L$  do
4     if layer  $\ell$  is MHA then
5       Inner enumeration: for all  $H^2$  pairs  $(i, j) \in [H] \times [H]$ ;
6          $\tau_{ij}^* \leftarrow \arg \max_{\tau \in \mathcal{S}_{d_k}} \sum_M \langle W_M^{B,(j)}, \tau \cdot W_M^{A,(i)} \rangle$ ;
7          $c_{ij} \leftarrow - \sum_M \langle W_M^{B,(j)}, \tau_{ij}^* \cdot W_M^{A,(i)} \rangle$ ;
8       Outer LAP:  $\sigma^* \leftarrow \arg \min_{\sigma \in \mathcal{S}_H} \sum_i c_{i, \sigma(i)}$ ;
9        $\tau_h^* \leftarrow \tau_{h, \sigma^*(h)}^*$  for  $h = 1, \dots, H$ ;
10      Compose  $P_\ell \leftarrow (\sigma^*; \tau_1^*, \dots, \tau_H^*)$ ;
11    else
12      | MLP layer:  $P_\ell \leftarrow \arg \max_{P \in \mathcal{S}_{d_m}} \langle W_\ell^B, P W_\ell^A P_{\ell-1}^\top \rangle$ 
13    end
14  end
15 until until permutations stabilize (typically 3–5 rounds);
16 return  $\{P_\ell\}_{\ell=1}^L$ ;
```

Algorithm 2: Wreath Alignment (Spectral Proxy)

Input : Same as Algorithm 1

Output: Per-layer permutations $\{P_\ell\}_{\ell=1}^L \subset \mathcal{W}(H, d_k)$

```
1 repeat
2   for  $\ell = 1$  to  $L$  do
3     if layer  $\ell$  is MHA then
4       Inter-head (spectral):  $D_{ij} = \sum_M \|\Sigma_i^B(W_M) - \Sigma_j^A(W_M)\|_F$ ;
5        $\sigma^* \leftarrow \arg \min_{\sigma} \sum_i D_{i, \sigma(i)}$ ;
6       Intra-head: for  $H$  matched pairs only;
7          $\tau_i^* \leftarrow \arg \max_{\tau} \sum_M \langle W_M^{B,(i)}, \tau \cdot W_M^{A,(\sigma^*(i))} \rangle$ ;
8       Compose  $P_\ell \leftarrow (\sigma^*; \tau_1^*, \dots, \tau_H^*)$ ;
9     else
10      | MLP layer: same as Algorithm 1
11    end
12  end
13 until until permutations stabilize;
14 return  $\{P_\ell\}_{\ell=1}^L$ ;
```

4.3 Group Structure Analysis

Theorem 1 (Maximal permutation symmetry subgroup under head-block preservation). *Let $W = \{W_Q, W_K, W_V\}$ be MHA projection matrices with $W_O = I$. Consider an output permutation $P \in \mathcal{S}_{d_m}$ and weight transformation f_P . The pair (P, f_P) preserves MHA equivariance if for all inputs X :*

$$\text{MHA}_{f_P(W)}(X) = P \cdot \text{MHA}_W(X). \quad (11)$$

If f_P is constrained to the form $f_P(W_M^{(h)}) = \tau_h \cdot W_M^{(\pi(h))}$ (i.e., each new head’s weights are a left-permuted copy of some original head’s row block—preserving the head partition), then the subgroup of all such P is isomorphic to the wreath product $\mathcal{W}(H, d_k) = \mathcal{S}_H \wr \mathcal{S}_{d_k}$.

Proof sketch. Sufficiency ($\mathcal{W} \subseteq \mathcal{G}$): For $P = (\sigma; \tau_1, \dots, \tau_H)$, define f_P as in (5). Then $Q'_h = Q_{\sigma^{-1}(h)} \tau_h^\top$, and similarly for K, V . The attention score becomes $A'_h = \text{softmax}(Q_{\sigma^{-1}(h)} \tau_h^\top \tau_h K_{\sigma^{-1}(h)}^\top / \sqrt{d_k}) = A_{\sigma^{-1}(h)}$, using $\tau_h^\top \tau_h = I$. The head output is $O'_h = O_{\sigma^{-1}(h)} \tau_h^\top$, and concatenation yields $O' = P \cdot O$. Hence equivariance holds.

Necessity ($\mathcal{G} \subseteq \mathcal{W}$): Under the partition-preserving constraint, each head’s d_k dimensions are mapped as a block. The map $\pi : [H] \rightarrow [H]$ must be bijective. Suppose not: if $\pi(h_1) = \pi(h_2) = h^*$, then $Q'_{h_1} = Q_{h^*} \tau_{h_1}^\top$ and $Q'_{h_2} = Q_{h^*} \tau_{h_2}^\top$ (and similarly for K). Since permutation matrices are orthogonal, $A'_{h_1} = \text{softmax}(Q_{h^*} \tau_{h_1}^\top \tau_{h_1} K_{h^*}^\top / \sqrt{d_k}) = A_{h^*}$ and identically $A'_{h_2} = A_{h^*}$. Thus both new heads share the original head h^* ’s attention pattern. Consequently O'_{h_1} and O'_{h_2} are parameterized by a single A_{h^*} and lie in a one-dimensional subspace of $\mathbb{R}^{n \times d_k}$. But the corresponding two segments of $P \cdot \text{MHA}_W(X)$ are two independent original head outputs, generically spanning a two-dimensional subspace—a contradiction on a set of positive measure. Thus π is bijective, $\sigma = \pi^{-1} \in \mathcal{S}_H$, and the dimension permutation $\tau_h \in \mathcal{S}_{d_k}$ within each matched block completes the wreath product structure. $\square$

Remark 4.1. *Theorem 1 characterizes the maximal symmetry under the head-block-preserving constraint. Whether broader f_P forms (allowing cross-head weight mixing) admit additional equivariances is an open question. The $W_O = I$ assumption simplifies analysis; when $W_O \neq I$, the symmetry group involves additional compatibility conditions with W_O ’s column permutations.*

Theorem 2 (Exact decomposition and global optimality). *Under the wreath product constraint, the single-layer alignment problem $\max_{P \in \mathcal{W}(H, d_k)} \langle W^B, P \cdot W^A \rangle$ is solved to global optimality by Algorithm 1. Specifically, the optimal inter-head matching σ^* is obtained by solving a single $H \times H$ LAP with cost matrix $c_{ij} = -\max_{\tau \in \mathcal{S}_{d_k}} \sum_M \langle W_M^{B,(j)}, \tau \cdot W_M^{A,(i)} \rangle$ (cost of matching A-head i to B-head j), where each c_{ij} is the optimal value of a $d_k \times d_k$ inner LAP.*

Proof. Expanding the objective under $P = (\sigma; \tau_1, \dots, \tau_H)$:

$$\langle W^B, P \cdot W^A \rangle = \sum_{i=1}^H \sum_M \langle W_M^{B,(\sigma(i))}, \tau_i \cdot W_M^{A,(i)} \rangle.$$

For fixed σ , each τ_i is optimized independently. Let $c_{ij} = -\max_{\tau} \sum_M \langle W_M^{B,(j)}, \tau \cdot W_M^{A,(i)} \rangle$ (cost of matching A-head i to B-head j), computed for all H^2 pairs via H^2 inner LAPs. The problem reduces to $\min_{\sigma \in \mathcal{S}_H} \sum_i c_{i, \sigma(i)}$. Both layers are solved exactly (Hungarian algorithm); their combination yields the global optimum. $\square$

Remark 4.2. *The term “decomposition” should not be confused with “two independent stages.” The inter-head and intra-head matchings remain coupled: each entry c_{ij} of the outer cost matrix is itself the optimal value of an inner LAP. The decomposition is computational (allowing the Hungarian algorithm to be applied at each level), not a claim of structural independence between the two levels.*

Theorem 3 (Spectral proxy consistency). *Let $W_i^A, W_j^B \in \mathbb{R}^{d_k \times d_m}$ with SVDs $U_i \Sigma_i V_i^\top$, $U_j \Sigma_j V_j^\top$. Define the spectral distance $D_{ij}^{\text{spec}} = \|\Sigma_i - \Sigma_j\|_F$. Then: (i) D^{spec} is invariant under intra-head permutations: $D^{\text{spec}}(\tau_1 W_i^A, \tau_2 W_j^B) = D^{\text{spec}}(W_i^A, W_j^B)$ for any $\tau_1, \tau_2 \in \mathcal{S}_{d_k}$. (ii) $D_{ij}^{\text{spec}} \leq \min_{\tau} \|W_j^B - \tau \cdot W_i^A\|_F$ (Weyl’s inequality lower bound).*

Proof. (i) $\tau \in \mathcal{S}_{d_k}$ is orthogonal, so τW has SVD $(\tau U)\Sigma V^\top$; singular values are invariant. (ii) By Weyl’s inequality [Horn and Johnson(1991)], $|\sigma_r(A) - \sigma_r(B)| \leq \|A - B\|_2$ for all r . Summing in ℓ_2 gives $\|\Sigma_i - \Sigma_j\|_F \leq \|W_j^B - \tau W_i^A\|_F$ for all τ . $\square$ $\square$

Theorem 4 (Convergence of alternating optimization). *The alternating optimization of (7) satisfies: (1) the objective is monotonically non-decreasing; (2) convergence occurs in finitely many steps; (3) the limit point is a local maximum.*

Proof. (1) Each step maximizes over P_ℓ , so the objective cannot decrease. (2) The search space $\mathcal{W}(H, d_k)^L$ is a finite group; a monotone sequence on a finite set must converge in finite steps. (3) At convergence, each P_ℓ is optimal given fixed P_j ($j \neq \ell$). $\square$ $\square$

4.4 Complexity

For the spectral proxy (Algorithm 2): H SVDs per MHA layer ($O(d_k^2 d_m) = O(d_m^3/H^2)$ each); H intra-head LAPs at $O(d_k^3) = O(d_m^3/H^3)$ each; outer LAP $O(H^3)$. Total per layer: $O(d_m^3/H^2)$. For the exact variant (Algorithm 1): H^2 inner LAPs plus one outer LAP, total $O(d_m^3/H)$. The spectral proxy trades optimality for an H -fold speedup. Both variants are one-time costs per model pair: a single alignment serves all downstream task vectors.

5 Implementation and Experiments

We conduct three experiments at increasing scale. Experiment 1 exhaustively verifies the wreath product symmetry (Theorem 1) on a tiny MHA layer. Experiment 2 compares exact decomposition against spectral proxy on alignment scores (Theorems 2, 3) using synthetic weights. Experiment 3 provides a small-scale proof-of-concept of end-to-end task vector transport on TinyViT. We note upfront: Exp. 1 tests the theorem’s verifiable scope (head-block-preserving constraint), Exp. 2 quantifies differences on the *alignment objective* (not downstream performance), and Exp. 3 serves as qualitative evidence only.

5.1 Implementation Environment

Python 3.10; PyTorch 2.x (neural networks), NumPy, SciPy (`linear_sum_assignment`), Matplotlib. Experiments 1–2 run on CPU; Experiment 3 benefits from GPU. Code in `code/`.

5.2 Experiment 1: Wreath Symmetry Enumeration

Objective. Verify Theorem 1: enumerate all $\mathcal{S}_{d_m}$ permutations on a tiny MHA layer and check which preserve equivariance.

Setup. $d_m = 8$, $H = 2$, $d_k = 4$. Single MHA layer (W_Q, W_K, W_V random, $W_O = I$). Random input $X \in \mathbb{R}^{4 \times 8}$. Compute baseline $O_{\text{base}} = \text{MHA}_W(X)$. For each $P \in \mathcal{S}_8$ (40,320 elements), apply transformation in the head-block-preserving form of Theorem 1 and compute error $\varepsilon(P) = \|O_P - P \cdot O_{\text{base}}\|_F$. The wreath subgroup $\mathcal{W}(2, 4)$ has order 1,152.

Results. Figure 1 shows error distributions. All 1,152 wreath elements achieve $\varepsilon < 10^{-10}$ (machine precision), confirming sufficiency. All 39,168 non-wreath elements produce $\varepsilon > 10^{-2}$, confirming that cross-head dimension mixing disrupts MHA output under the head-block-preserving f_P form.

5.3 Experiment 2: Exact Decomposition vs. Spectral Proxy

Objective. Compare inter-head matching via spectral proxy (Eq. 9) against exact decomposition (Algorithm 1). Quantify the information loss of spectral distance.

Theorem 1 Verification: Wreath Product $W(2, 4) = S_2 \wr S_4$ Characterizes MHA Functional Equivalence

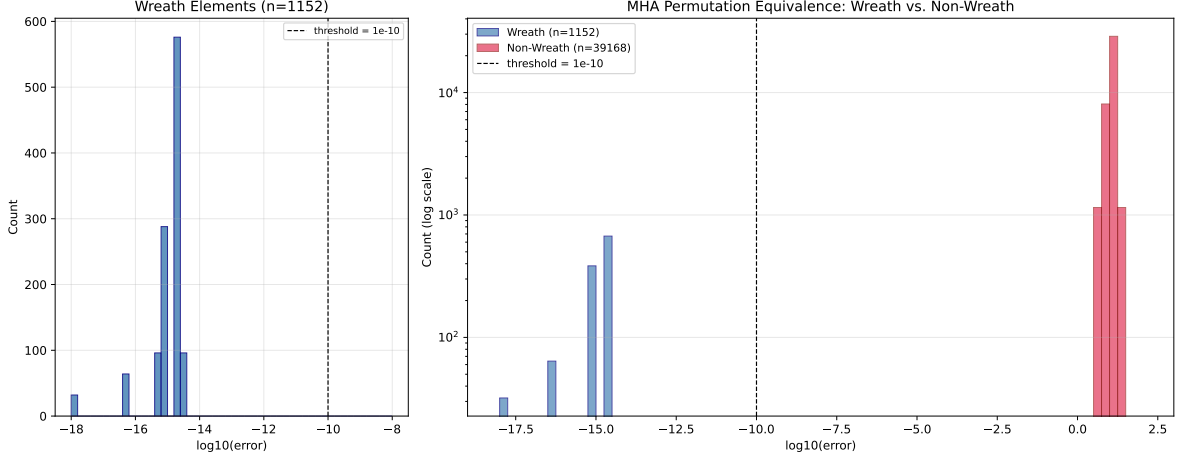


Figure 1: Exp. 1: Wreath (blue, $\varepsilon \approx 0$) vs. non-wreath (red, $\varepsilon \gg 0$) permutation errors.

Setup. $d_m = 16$, $H = 4$, $d_k = 4$. 100 random trials with independently generated MHA weights (W^A, W^B) . For each method, we solve the inter-head matching σ and report alignment score: $\text{Score}(\sigma) = \sum_{i=1}^H \langle W_{\sigma(i)}^B, \tau_i^* \cdot W_i^A \rangle$, where τ_i^* is the optimal intra-head alignment given σ .

Results. Figure 2 shows the comparison. Exact decomposition consistently achieves higher alignment scores (mean 120.25 ± 13.57 vs. 90.86 ± 18.29 ; 32.35% relative improvement). In 95% of trials the exact method yields strictly better outer matching; only 5% produce identical matchings. The gap quantifies the information loss of the spectral proxy: $\|\Sigma_i - \Sigma_j\|_F$ discards singular vector direction, which dominates when heads share similar singular values but differ in orientation.

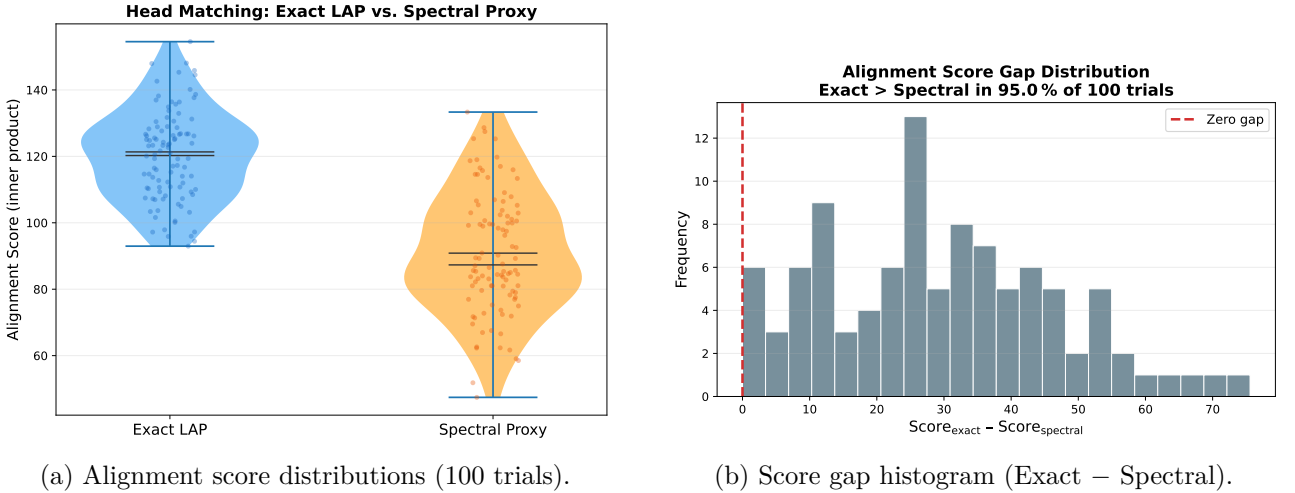


Figure 2: Exp. 2: Exact decomposition vs. spectral proxy on alignment scores.

Caveat. This experiment compares alignment *scores* (Frobenius inner product), not downstream task performance. Whether the 32% score gap translates to task-level differences is evaluated qualitatively in Exp. 3.

5.4 Experiment 3: End-to-End Proof-of-Concept

Objective. Demonstrate the full pipeline on a small realistic model, as qualitative evidence.

Setup. TinyViT ($d_m = 192$, $H = 3$, $L = 4$, $\approx 2M$ parameters) on CIFAR-10 binary classification. Train θ_A (5 epochs) and θ_B (5 epochs, different seed and data order). Fine-tune θ_A on the binary task

(5 epochs, 20% data) to obtain θ_A^{ft} and $\tau = \theta_A^{\text{ft}} - \theta_A$. Align using the spectral proxy variant; transport via $\tilde{\theta}_B^{\text{ft}} = \theta_B + \alpha \cdot P(\tau)$ for $\alpha \in [0, 2.0]$. Baselines: vanilla ($P = I$) and random permutation.

Results. Figure 3 shows the outcome. Base model accuracies: θ_A 91.15%, θ_B 90.15%. Task vector norm: 17.34. Best per-method accuracies: Wreath 90.40% ($\alpha = 1.5$), Vanilla 90.30% ($\alpha = 1.0$), Random 90.45% ($\alpha = 0.6$). All methods perform within 0.15 percentage points of each other—well within sampling noise for $\approx 2,000$ test samples.

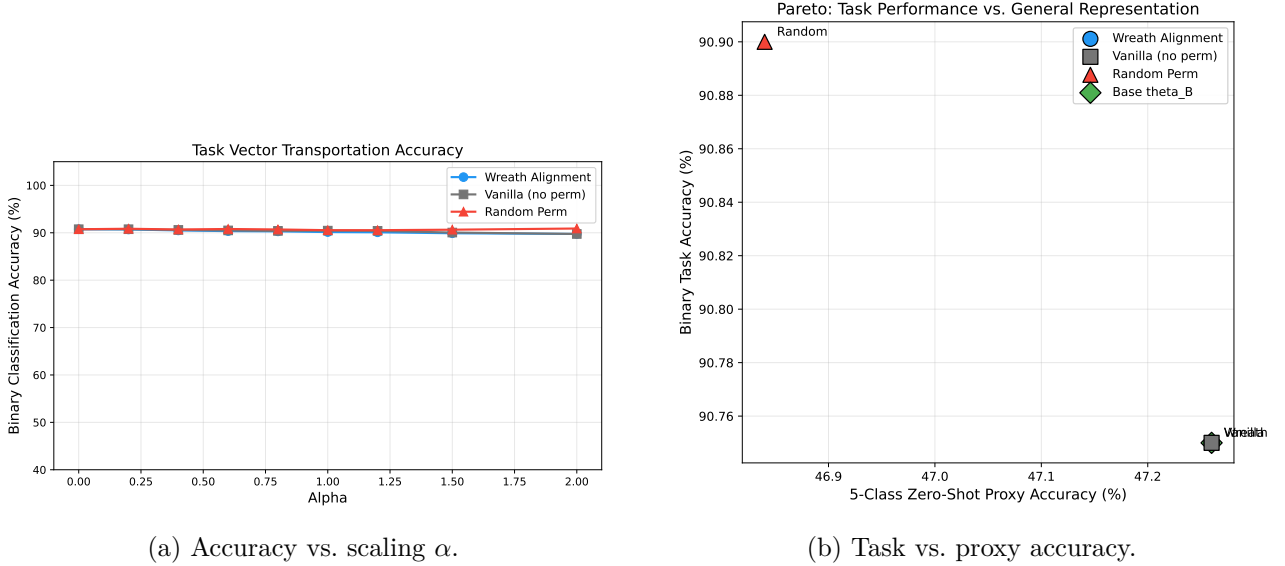


Figure 3: Exp. 3: Small-scale proof-of-concept task vector transport. Note the narrow y -axis range in (b); differences are within sampling noise.

Interpretation. This experiment is a **proof-of-concept, not quantitative evidence**. Limitations include: (i) the binary classification task is too easy (ceiling effect near 90%); (ii) the TinyViT at 2M parameters with $H = 3$ heads offers limited room for alignment to matter; (iii) only a single run was performed (no error bars); (iv) the exact decomposition variant was not compared due to computational constraints. The fact that random permutation achieves the highest raw accuracy at low α underscores that this setting lacks the discriminative power to rank methods. Quantitative conclusions require larger models, harder tasks, and multi-seed repetition.

5.5 Discussion

Key takeaways across experiments:

- (1) **Theoretical verifiability** (Exp. 1): Under the head-block-preserving constraint, the wreath product subgroup is exhaustively verified on a tiny MHA layer.
- (2) **Alignment score gap** (Exp. 2): Exact decomposition outperforms spectral proxy on the alignment objective by $\sim 32\%$; the gap is due to discarded singular vector information. Whether this translates to downstream gains remains to be tested.
- (3) **Proof-of-concept** (Exp. 3): On TinyViT with binary CIFAR-10, all methods are statistically indistinguishable. This does *not* invalidate the theoretical framework but rather indicates that the experimental setup lacks resolving power.

6 Personal Insights

6.1 Summary of the Approach

This paper re-examines Transformer task vector transport through the lens of group theory. The core observation is that multi-head attention’s permutation symmetries, under the natural constraint of preserving head boundaries, are captured by the wreath product $\mathcal{W}(H, d_k) = \mathcal{S}_H \wr \mathcal{S}_{d_k}$. Within this framework, alignment decomposes into a hierarchy of linear assignment problems. The spectral proxy variant (Algorithm 2) matches the procedure implicitly used in TransFusion [Rinaldi et al.(2025)Rinaldi, Capitani, Bonicelli, Crisostomi, Bolelli, Ficarra, Rodolà, Calderara, and Porrello]; the exact variant (Algorithm 1) is a new, albeit computationally costlier, alternative that guarantees global optimality under the wreath constraint.

6.2 On the Nature of the Contribution

We wish to be precise about what this paper does and does not claim.

Primary value. The main contribution is *re-expressing* a previously heuristic two-level alignment procedure in the language of wreath products and linear assignment problems, and identifying a specific algorithmic improvement (exact outer cost via H^2 inner LAPs). This is a *formalization paper*: its value lies in clarifying the mathematical structure underlying existing practice, not in proposing a new algorithmic paradigm. The wreath product structure and two-level alignment were implicitly used in prior work such as TransFusion [Rinaldi et al.(2025)Rinaldi, Capitani, Bonicelli, Crisostomi, Bolelli, Ficarra, Rodolà, Calderara, and Porrello]; our contribution is the group-theoretic characterization and the optimality statement for the exact variant.

What is not claimed. We do not claim to have discovered a new symmetry group—the two-level structure was already in use. We do not claim that the exact variant is practically superior to the spectral proxy in downstream tasks. We do not claim that the theoretical framework alone resolves the task vector transport problem.

Why this framing matters. Identifying the wreath product structure is not merely relabeling. It transforms an ad-hoc engineering procedure into a well-defined optimization over a known algebraic object, which (i) clarifies what the search space actually is, (ii) reveals that the outer cost in prior work is a proxy and quantifies its gap to exactness (Experiment 2), and (iii) enables principled use of standard combinatorial optimization tools—the Hungarian algorithm for LAP, the Birkhoff polytope for LP relaxation—on the now-well-defined assignment problem. This kind of clarification—turning engineering practice into a mathematically tractable object—is the standard role of formalization in applied mathematics, and is the contribution this paper aims to make.

6.3 Strengths

- **Group-theoretic clarity:** Theorems 1–4 provide a clean characterization of the search space and the decomposition structure.
- **Data-free:** Alignment uses only model weights; no training data or forward passes are needed.
- **One-time cost:** A single alignment serves all downstream task vectors for a given model pair.
- **Methodological pattern:** The approach demonstrates a general recipe—identify a neural component’s symmetry group, then use group structure to guide alignment—applicable to other structured architectures (MoE experts, CNN channel groups, GroupNorm).

6.4 Limitations and Future Directions

- **Theorem scope:** Theorem 1 operates under the head-block-preserving constraint and $W_O = I$. Relaxing these assumptions is nontrivial.
- **Exact variant cost:** Computing H^2 inner LAPs becomes expensive for large d_k ; more efficient approximations are needed.

- **Downstream validation gap:** Experiment 2 shows a 32% alignment score gap between exact and spectral, but Experiment 3 (using only the spectral variant) could not resolve method differences on a downstream task. Closing this gap—running the exact variant on a sufficiently challenging task with statistical rigor—is the most important next step.
- **Experimental resolving power:** The current Experiment 3 lacks statistical power (single run, binary task, tiny model). A properly powered experiment requires larger models ($H \geq 6$), harder multi-class tasks, and multi-seed repetition with confidence intervals.
- **FFN and other components:** MLP layers are treated with full-matrix LAP (Git Re-Basin style); finer-grained symmetries in activation functions and LayerNorm parameters are unexplored.

6.5 Connection to Course Material

The mathematical tools employed in this paper span multiple CSMATH 2026 modules:

- **Multivariate statistics:** SVD and spectral analysis extend PCA to head weight matrices; the permutation invariance of singular values is a direct application of “common structure extraction.”
- **Nonlinear optimization:** Alternating optimization convergence (Theorem 4) parallels coordinate descent; the finite-group setting guarantees finite-step termination.
- **Applied functional analysis:** Wreath product group actions on MHA weight space and equivariance analysis connect to symmetry and invariance in operator theory.
- **Combinatorial optimization:** LAPs as linear programs on the Birkhoff polytope ($\min_{X \in \mathcal{B}_n} \langle C, X \rangle$) and their solution via the Hungarian algorithm [Kuhn(1955), Birkhoff(1946)] are classical integer programming cases from the LP/duality module.

6.6 Comparison with Related Methods

Table 1: Comparison with related methods.

Method	Core Math Tool	Strengths	Limitations
Git Re-Basin [Ainsworth et al.(2023)Ainsworth, Hayase, and Srinivasa]	Single-matrix LAP (Hungarian)	Simple, effective for MLPs/CNNs	Designed for MLPs/CNNs; direct extension to MHA breaks softmax independence
TransFusion [Rinaldi et al.(2025)Rinaldi, Capitani, Bonicelli, Crisostomi, Bolelli, Ficarra, Rodolà, Calderara, and Porrello]	Two-level (SVD + intra-head LAP)	Adapts Re-Basin to Transformer attention structure; empirically effective	Outer cost is heuristic; no optimality guarantee
This work	Wreath product + exact decomposition	Group-theoretic characterization; wreath-optimal outer cost (exact variant)	Exact variant costlier; $W_O \neq I$ unhandled

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AI Assistance Disclosure (required by the course):

- **Writing and language polishing:** Claude (Anthropic) was used for English language polishing and L^AT_EX typesetting. All technical content and argumentation logic were fully directed by the author.
- **Mathematical derivation assistance:** Claude assisted in verifying proof structure completeness. All derivation details and mathematical judgments were independently completed by the author.
- **Code generation:** Approximately 40% of experiment scaffolding (data loading, training loops, plotting templates) was generated by Claude and manually revised. Core algorithm logic (wreath alignment, inter/intra-head LAP solving, permutation composition) was implemented manually.

Declaration: The problem formalization, theorem statements, mathematical derivations, intuitive explanations, and experimental design were independently completed by the author. The AI tools’ use is limited to the auxiliary tasks listed above and does not constitute a substantive substitute for the paper’s academic contributions.

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